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Critical roles of chemokine receptor CCR10 in regulating memory IgA responses in intestines

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Chemokine receptor CCR10 is expressed by all intestinal IgA-producing plasma cells and is suggested to play an important role in positioning these cells in the lamina propria for proper IgA production to maintain intestinal homeostasis and protect against infection. However, interfering with CCR10 or its ligand did not impair intestinal IgA production under homeostatic conditions or during infection, and the in vivo function of CCR10 in the intestinal IgA response remains unknown. We found that an enhanced generation of IgA$^+$ cells in isolated lymphoid follicles of intestines offset defective intestinal migration of IgA$^+$ cells in CCR10-KO mice, resulting in the apparently normal IgA production under homeostatic conditions and in primary response to pathogen infection. However, the compensatorily generated IgA$^+$ cells in CCR10-KO mice carried fewer hypermutations in their Ig heavy chain alleles than those of WT mice, indicating that their IgA repertoires are qualitatively different, which might impact the intestinal homeostasis of microflora. In addition, CCR10-deficient long-lived IgA-producing plasma cells and IgA$^+$ memory B cells generated against the pathogen infection could not be maintained properly in intestines. Consequently, IgA memory responses to the pathogen reinfection were severely impaired in CCR10-KO mice. These findings elucidate critical roles of CCR10 in regulating the intestinal IgA response and memory maintenance and could help in design of vaccines against intestinal and possibly other mucosal pathogens.

CCL28 | citrobacter | gut-homing | T cell-dependent response | T cell-independent response

IgA antibodies are important components of the mucosal immune system. In the intestine, they are produced by IgA-producing plasma cells predominantly localized in the lamina propria (LP) and secreted into the lumen, where they play important roles in maintaining homeostasis of commensal microflora and neutralizing food-borne pathogens and toxins (1). In normal mice, the majority of intestinal IgA-producing plasma cells originate in Peyer patches from naive B cells in response to the stimulation of intestinal antigens (2). After undergoing the isotype switch and affinity maturation in Peyer patches, IgA$^+$ plasmablast cells migrate into effector sites such as the intestinal LP, where they differentiate further to become mature IgA-producing plasma cells. IgA-producing plasma cells are also generated in isolated lymphoid follicles (ILFs), the small follicles composed predominantly of B cells and scattered abundantly in small and large intestines of humans and mice (2–5). ILFs are formed only after colonization of commensal bacteria in the intestines, suggesting that they are involved in regulating the intestinal homeostasis of microflora (6). It was also reported that the intestinal LP itself could be a site for the in situ generation of IgA-producing plasma cells (7). These processes cooperate to maintain proper generation of IgA antibodies in the intestine.

Although generation of the intestinal IgA-producing plasma cells is studied extensively, molecular mechanisms regulating the IgA maintenance and memory responses are still poorly understood. It was reported recently that the maintenance of intestinal IgA production is significantly different from that of systemic IgG production (8). It was found that, in a germ-free condition, antigen-specific IgA$^+$ plasma cells could be maintained in the intestine for a long time in the absence of antigenic stimulation (half-life > 16 wk), suggesting that unique intrinsic properties of the IgA-producing plasma cells and intestinal environments might collaborate to maintain the prolonged IgA production. However, maintenance of the antigen-specific IgA-producing plasma cells is significantly affected by continuous presence of commensal bacteria, which induce generation of new IgA-producing plasma cells that replace the existing antigen-specific IgA$^+$ cells in the intestine. Molecular factors involved in the long-term IgA maintenance are largely unknown and it is also not well understood how IgA memory responses to pathogen infection are regulated in the intestines.

Chemokine receptor CCR10 is expressed on nearly all IgA$^+$ plasma cells and is suggested to play an important role in directing migration of the IgA$^+$ cells generated in Peyer patches or other mucosa-associated lymphoid tissues into effector sites such as the intestinal LP through interaction with its mucosa-specific ligand CCL28 expressed by intestinal epithelial cells (9–12). Consistent with this notion, intestinal, but not systemic, immunization of humans efficiently generated CCR10$^+$ antigen-specific IgA$^+$ cells (13). In addition, most human bloodborne IgA$^+$ plasma cells express CCR10, suggesting that they originate from mucosal responses (14).

Despite the multiple lines of evidence implicating CCR10 in the intestinal IgA response, the role of CCR10 in this process is not clear. One earlier study reported that neutralizing the CCR10 ligand CCL28 with antibodies impaired intestinal IgA production in response to oral immunization of cholera toxin (CT) in a mouse model (9). However, the anti-CCL28 antibody treatment did not have any effect on the IgA response to intestinal rotavirus infection (15). More directly, there was no defect of homeostatic IgA production in intestines of CCR10-KO mice (16). These studies suggest that CCR10/ligands are not critically required for normal levels of IgA responses to commensal bacteria or pathogen infection, and the functional importance of CCR10 in the intestinal IgA response is still unclear. By using a strain of CCR10-KO/EGFP-knock-in mice (17), we investigated expression and involvement of CCR10 in intestinal IgA responses under homeostatic conditions and during bacterial pathogen infection. In this report, we provide definite evidence that CCR10 is critical in the intestinal IgA response and memory maintenance.

Results

Defective Migration of CCR10-Deficient IgA$^+$ Plasma Cells into Small and Large Intestines. To understand how CCR10 is involved in regulating IgA responses in the intestine, we used a strain of

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CCR10-KO/EGFP-knock-in mice in which the coding region of enhanced GFP (EGFP) replaced most of the CCR10 coding region from its translation start site down, allowing us to use the knocked-in EGFP as a reporter for endogenous CCR10 expression (17). First, we determined effects of CCR10 KO on intestinal IgA production under normal specific pathogen-free conditions. In heterozygous CCR10-KO/EGFP-knock-in ($CCR10^{+/EGFP}$) mice, nearly all mature IgA^+ plasma cells ($B220^-CD19^-$) of intestines were $EGFP^+$ (Fig. 1*A*), consistent with the previous report that they expressed CCR10 (9). Intestinal IgA^+ plasma cells of homozygous $CCR10^{EGFP/EGFP}$ were also $EGFP^+$, and their numbers were not different from those of WT ($CCR10^{+/+}$) or $CCR10^{+/EGFP}$ mice (Fig. 1*A*). Mean fluorescence intensities of EGFP signals in the IgA^+ cells of $CCR10^{EGFP/EGFP}$ mice were two times those of $CCR10^{+/EGFP}$ mice (Fig. 1*A*), suggesting that both alleles were coexpressed and the knocked-in EGFP could reliably report CCR10 expression. Levels of total IgA antibodies in feces were similar in $CCR10^{EGFP/EGFP}$, $CCR10^{+/EGFP}$, and WT mice (Fig. 1*B*), consistent with the normal numbers of intestinal IgA^+ plasma cells and results of a recent report (16). The IgA^+ plasma cells of $CCR10^{EGFP/EGFP}$ and control mice also had the similar capacity to produce IgA in vitro (Fig. S1).

Because no other chemokine receptor could substitute for the CCR10-mediated migration of IgA^+ cells toward its mucosal ligand CCL28 in vitro (16), it is unclear why the CCR10 KO did not have much effect, particularly in the large intestine, where the ligand (CCL25) for another major gut-homing molecule, CCR9, is not expressed (9). We then tested whether CCR10-deficient IgA plasma cells were defective in migration into intestines by using an in vivo migration assay in which similar numbers of $EGFP^+IgA^+$ cells of $CCR10^{+/EGFP}$ and $CCR10^{EGFP/EGFP}$ mice were cotransferred into WT mice. Notably, significantly fewer $EGFP^+IgA^+$ cells of the $CCR10^{EGFP/EGFP}$ donor ($CD45.1^-CD45.2^+$) were recovered from large and small intestines of the recipients than those of the $CCR10^{+/EGFP}$ donor ($CD45.1^+CD45.2^+$) 60 h after the transfer (Fig. 1 *C* and *D*). These results suggest that CCR10 is required for efficient migration of IgA^+ cells into the intestines and no other molecule could fully compensate for its function in this process. Therefore, the apparently normal levels of intestinal IgA^+ cells and antibodies in $CCR10^{EGFP/EGFP}$ mice result from a compensation mechanism other than functional redundancy of any other homing receptor with CCR10.

Enhanced Generation of IgA^+ Cells Within Increased Numbers of ILFs in $CCR10^{EGFP/EGFP}$ Mice. One possible mechanism to compensate for the impaired intestinal localization of IgA^+ cells in $CCR10^{EGFP/EGFP}$ mice is to increase their generation. ILFs, the dynamic B-cell–rich follicles in intestines, are likely a site for the compensational generation of IgA^+ cells. Supporting this notion, there were significantly more ILFs in intestines of $CCR10^{EGFP/EGFP}$ mice than of $CCR10^{+/EGFP}$ mice based on an en face whole-mount staining method that identifies clusters of $B220^+$ cells of the ILFs (Fig. 2*A*) (18, 19). Consistent with the increased numbers of ILFs, there were higher percentages of $B220^+$ B cells in lymphocytes isolated from LP of intestines (LPLs) of $CCR10^{EGFP/EGFP}$ than of $CCR10^{+/EGFP}$ mice, whereas percentages of $CD3^+$ T cells were similar (Fig. 2*B* and Fig. S2*A*). Percentages of $B220^+IgA^+$ plasmablasts were also higher in the LPLs of $CCR10^{EGFP/EGFP}$ than of $CCR10^{+/EGFP}$ mice, consistent with the enhanced generation of IgA^+ cells (Fig. 2*B*). Most of the $B220^+IgA^+$ cells were $EGFP^-$ (Fig. S2*B*), suggesting that CCR10 is up-regulated in a later phase of the IgA^+ cell differentiation. Immunofluorescent staining of

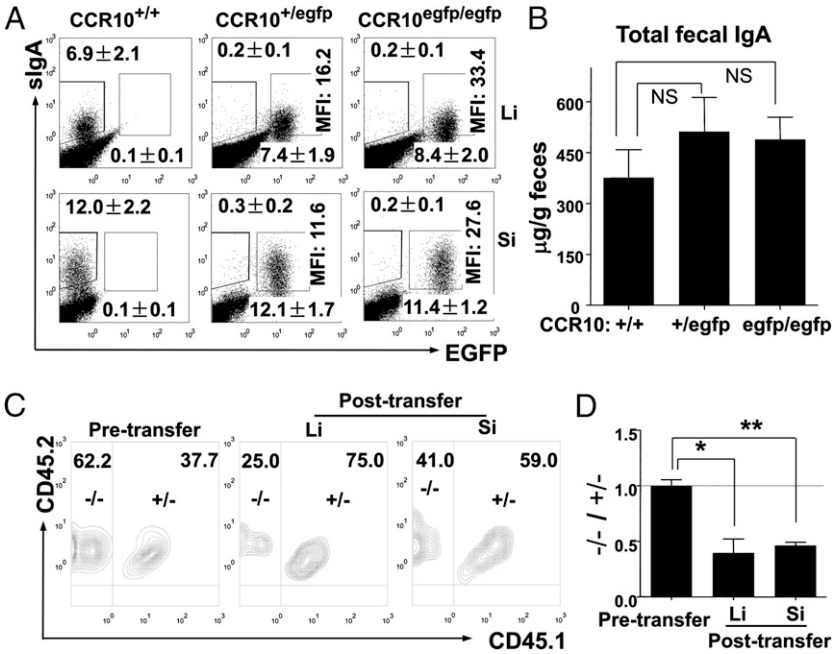

Fig. 1. CCR10-deficient IgA^+ cells are defective in migration into intestines in vivo. (*A*) Flow cytometry (i.e., FACS) analysis of the LPLs isolated from large intestine (Li) and small intestine (Si) of $CCR10^{EGFP/EGFP}$, $CCR10^{+/EGFP}$, and WT mice for detection of surface IgA ($sIgA^+$) plasma cells and their expression of EGFP (CCR10). Gated on $CD3^-B220^-CD19^-$ cells: the numbers next to each gate represent the mean percentage [or mean fluorescence intensity (MFI) of EGFP] and SEM of the gated cells; $n = 7$ (large intestine) and 4 (small intestine). (*B*) Amounts of total IgA antibodies in feces of $CCR10^{EGFP/EGFP}$ and control mice, as determined by ELISA; $n = 6$–7 per group. For all figures: NS, not significantly different; $*P < 0.05$, $**P < 0.01$, and $***P < 0.001$. (*C* and *D*) Poor recovery of the $CCR10^{EGFP/EGFP}$ donor-derived $EGFP^+sIgA^+$ cells in both small and large intestines of WT recipients adoptively transferred with a mixture of $EGFP^+sIgA^+$ cells of $CCR10^{+/EGFP}$ and $CCR10^{EGFP/EGFP}$ mice. Representative FACS histographs of the mixed $EGFP^+sIgA^+$ cells of $CCR10^{+/EGFP}$ ($CD45.1^+CD45.2^+$) and $CCR10^{EGFP/EGFP}$ ($CD45.1^-D45.2^+$) mice injected into (pretransfer) and recovered from (posttransfer) the recipients (*C*), in which the values indicate percentages of $EGFP^+sIgA^+$ cells of $CCR10^{+/EGFP}$ ($^{+/-}$) and $CCR10^{EGFP/EGFP}$ ($^{-/-}$) origins. Mean ± SEM of relative ratios of the $CCR10^{EGFP/EGFP}$ vs. $CCR10^{+/EGFP}$ derived $EGFP^+sIgA^+$ cells recovered from the recipients in three independent experiments (*D*), with the ratio set as 1:1 for the pretransfer mixture.

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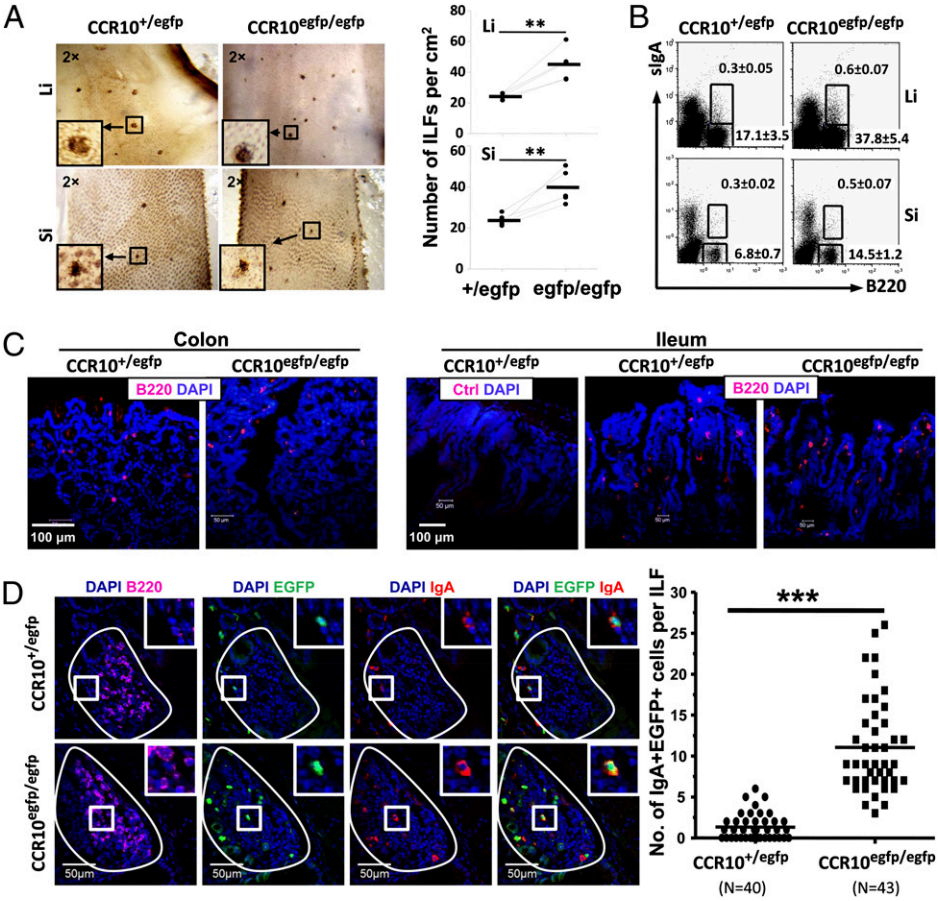

Fig. 2. Enhanced generation of IgA$^+$ cells within increased numbers of ILFs in CCR10-KO mice. (*A*) Densities of ILFs in large intestine (Li) and small intestine (Si) of CCR10$^{EGFP/EGFP}$ and CCR10$^{+/EGFP}$ mice. ILFs were visualized based on the B220$^+$ clusters (dark brown) of the whole-mount staining of intestines and enumerated (left two panels). (*Inserts*) Amplified images of single ILFs. Numbers of ILFs per surface of intestines (in cm^2) are shown on the right. Each dot represents the result from one mouse. Results of paired mice analyzed in same experiments are linked with lines. The short flat lines indicate mean values of the groups. (*B*) FACS analysis of the LP lymphocytes of CCR10$^{EGFP/EGFP}$ and CCR10$^{+/EGFP}$ mice stained for sIgA and B220. Gated on CD3$^-$ cells: n = 7 (large intestine), n = 3 (small intestine). (*C*) Representative fluorescent microscopic images of cryosections of colon and ileum of CCR10$^{EGFP/EGFP}$ and CCR10$^{+/EGFP}$ mice to detect B220$^+$ cells in the LP region. An isotype-control antibody staining (ctrl) is shown (*Center*). n = 8–9. (*D*) Immunofluorescent microscopy of colonic cryosections of CCR10$^{EGFP/EGFP}$ and CCR10$^{+/EGFP}$ mice to detect IgA$^+$ cells in ILFs. Circled areas indicate ILFs that are distinguished from the surrounding LP areas by the densely packed B220$^+$ and other immune cells. The square insets (*Top Right*) are amplified images of the gated small squares. Numbers of EGFP$^+$IgA$^+$ cells per ILF are shown (*Right*), with one dot representing one ILF. Data are pooled from eight mice per group.

intestinal sections detected similar densities of B220$^+$ cells in the LP regions of CCR10$^{EGFP/EGFP}$ and CCR10$^{+/EGFP}$ mice (Fig. 2*C*), consistent with the notion that the higher percentages of B cells in intestines of CCR10$^{EGFP/EGFP}$ mice are mainly a result of their increased ILF numbers.

We then assessed numbers of IgA$^+$ cells in ILFs by immunofluorescent microscopy. Much more IgA$^+$ cells were found within ILFs of CCR10$^{EGFP/EGFP}$ than in those of CCR10$^{+/EGFP}$ mice (Fig. 2*D*). The majority of IgA$^+$ cells in the ILFs were EGFP$^+$, suggesting that they were meant to express CCR10. In contrast, the density of IgA$^+$ cells in the intestinal LP areas of CCR10$^{EGFP/EGFP}$ mice was similar to, if not slightly lower than, that in the CCR10$^{+/EGFP}$ controls (Fig. S3). Based on the increased numbers of ILFs and their enhanced generation of IgA$^+$ cells, we estimated that there were 11 times more IgA$^+$EGFP$^+$ plasma cells in ILFs of CCR10$^{EGFP/EGFP}$ mice than in those of CCR10$^{+/EGFP}$ controls (Fig. S4*A*). IgA-producing capacities were similar in ILF and LP IgA$^+$ plasma cells of CCR10$^{+/EGFP}$ and CCR10$^{EGFP/EGFP}$ mice (Fig. S4*B*). Although the increased IgA$^+$ plasma cells of ILFs account only for approximately 2% of IgA$^+$ plasma cells of LP in CCR10$^{EGFP/EGFP}$ mice, contribution of the ILF-originated IgA$^+$ plasma cells to the intestinal IgA production might be much greater than this number suggested, as the mature IgA$^+$ plasma cells generated in ILFs would likely migrate out continuously and end up in LP for a long-term production of IgA antibodies. Together, these results demonstrate that the impaired migration of CCR10-deficient IgA$^+$ cells into intestines is compensated for (at least in part) by their enhanced generation within increased numbers of ILFs. There were same numbers of Peyer patches and same percentages of IgA$^+$ cells within them in CCR10$^{EGFP/EGFP}$ and CCR10$^{+/EGFP}$ mice (Fig. S5), suggesting that they are not involved in the compensational generation of IgA$^+$ cells.

Reduced Hypermutation in IgA Antibodies of CCR10$^{EGFP/EGFP}$ Mice. ILFs support a T-cell–independent IgA isotype switch (3), which generates antibodies carrying fewer hypermutations than with a T-cell–dependent process. We therefore assessed mutation rates in the Ig heavy chain (IgH) alleles of intestinal IgA$^+$ cells of CCR10$^{EGFP/EGFP}$ mice. Compared with the WT controls, significantly higher percentages of the CCR10-KO IgA$^+$ cells carried germline sequences whereas much fewer of them carried high numbers of mutations (i.e., >5) in the assayed IgH region (Fig. 3*A*). On average, mutation rates were 8.75 and 4.56 nucleo-

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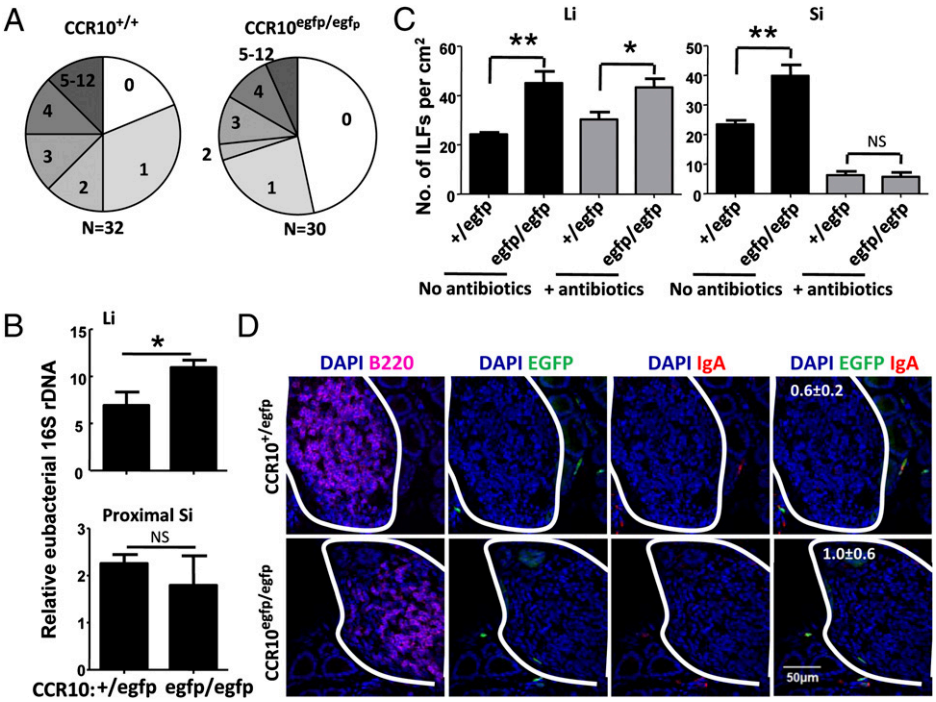

Fig. 3. Involvement of commensal bacteria in compensatory generation of IgA$^+$ cells in ILFs of CCR10$^{EGFP/EGFP}$ mice. (*A*) Reduced hypermutation in IgH alleles of intestinal EGFP$^+$IgA$^+$ cells of CCR10$^{EGFP/EGFP}$ mice. The hypermutation frequency was assessed in a 307-bp 3′ region franking the rearranged VHJ558/D/JH4 allele. The numbers in the charts indicate nucleotide mutations of the 307-bp-long sequence. Numbers of sequences analyzed are shown below the charts. (*B*) Relative burdens of commensal bacteria in large and proximal small intestines, calculated based on the quantitative PCR quantification of bacterial 16S rDNA ($n \geq 4$). (*C*) Numbers of ILFs in large and small intestines of CCR10$^{EGFP/EGFP}$ and CCR10$^{+/EGFP}$ mice 2 wk after antibiotic treatment. Numbers of ILFs of untreated mice are included as controls. (*D*) Representative fluorescent microscopy of colonic ILFs (circled) of CCR10$^{EGFP/EGFP}$ and CCR10$^{+/EGFP}$ mice 2 wk after antibiotic treatment. Average numbers (±SEM) of EGFP$^+$IgA$^+$ cells per ILF are shown (*Right*); n = 4 mice for each genotype.

tides per 1 kb in the CCR10$^{EGFP/EGFP}$ and CCR10$^{+/EGFP}$ samples, respectively (the basal mutation rate of this assay was 0.59 nucleotides per kb). Therefore, although production of IgA was compensated for in CCR10$^{EGFP/EGFP}$ mice, quality of the antibodies might be different from that of WT mice.

Involvement of Commensal Bacteria in Compensational Generation of IgA$^+$ Cells in ILFs of CCR10$^{EGFP/EGFP}$ Mice. The reduced IgA hypermutation could have impacts on intestinal homeostasis. Consistent with this, CCR10$^{EGFP/EGFP}$ mice had slightly higher loads of microflora in the mucosa of large intestines than CCR10$^{+/EGFP}$ mice (Fig. 3*B*), although compositions of several major groups of commensal bacteria were mostly normal, and titers of fecal IgA antibodies against them were not significantly affected (Fig. S6), suggesting a minor disturbance of intestinal homeostasis of the microflora. As development of ILFs and IgA isotype switch are induced by the colonization of commensal bacteria in intestines (3, 6), the higher levels of intestinal microflora in CCR10$^{EGFP/EGFP}$ mice are likely associated with the enhanced generation of ILFs and IgA. To test this, we treated the mice with a mixture of antibiotics at ages of 5 wk, when many ILFs are fully developed in the large but not small intestine (6). A 2-wk treatment drastically reduced development of ILFs in small intestines; ILF numbers become similar in the treated CCR10$^{EGFP/EGFP}$ and CCR10$^{+/EGFP}$ mice (Fig. 3*C*). On the contrary, the antibiotic treatment did not significantly reduce numbers of ILFs in colons; in which ILF numbers were still higher in the treated CCR10$^{EGFP/EGFP}$ than in CCR10$^{+/EGFP}$ mice (Fig. 3*C*). However, there were essentially no IgA$^+$ cells in the colonic ILFs of the treated CCR10$^{EGFP/EGFP}$ mice, the same as in the CCR10$^{+/EGFP}$ mice (Fig. 3*D*), indicating that the enhanced generation of IgA$^+$ cells within ILFs of CCR10$^{EGFP/EGFP}$ mice is stimulated by commensal bacteria and could be reduced by decreasing the commensal bacteria with antibiotic treatment. The antibiotics did not decrease titers of fecal IgA antibodies, even after a longer period of treatment (Fig. S7), likely because intestinal IgA$^+$ plasma cells are long-lasting (i.e., half-life of >16 wk) even in absence of stimulation from commensal bacteria (8).

Dysregulated Distribution of CCR10-Deficient IgA$^+$ Cells in Intestines and Internal Lymphoid Tissues. The compensation process made it difficulty to dissect how intrinsic expression of CCR10 on IgA$^+$ cells is important for their migration and maintenance in the intestines using the CCR10$^{EGFP/EGFP}$ mice. We then performed competitive bone marrow (BM) transfer experiments in which similar numbers of BM cells of CCR10$^{EGFP/EGFP}$ and CCR10$^{+/EGFP}$ mice were cotransferred into lethally irradiated WT mice. Markedly, whereas the donor cells of both types reconstituted BM of the recipients with similar efficiencies 7 to 8 wk after the transfer (Fig. 4*A*), IgA$^+$ cells of the CCR10$^{EGFP/EGFP}$ donor origin were profoundly under-represented in small and large intestines (Fig. 4*B*). In contrast, there were more IgA$^+$ cells of CCR10$^{EGFP/EGFP}$ than CCR10$^{+/EGFP}$ origin in internal lymphoid tissues such as spleens and BM (Fig. 4*B*). These results demonstrate that (*i*) intrinsic expression of CCR10 by IgA$^+$ cells is critically important for their efficient migration and/or maintenance in the intestines, and (*ii*), in its absence, the IgA$^+$ cells abnormally accumulate in the internal tissues. Further supporting the second notion, significantly higher percentages of EGFP$^+$IgA$^+$ cells were also found in BM, spleens, and mesenteric lymph nodes (MLNs) of unmanipulated CCR10$^{EGFP/EGFP}$ than of CCR10$^{+/EGFP}$ mice (Fig. 4*C* and Fig. S8). Serum levels of IgA were also slightly higher in CCR10$^{EGFP/EGFP}$ mice than in CCR10$^{+/EGFP}$ mice, whereas those of IgG, as controls, were similar (Fig. 4 *D*

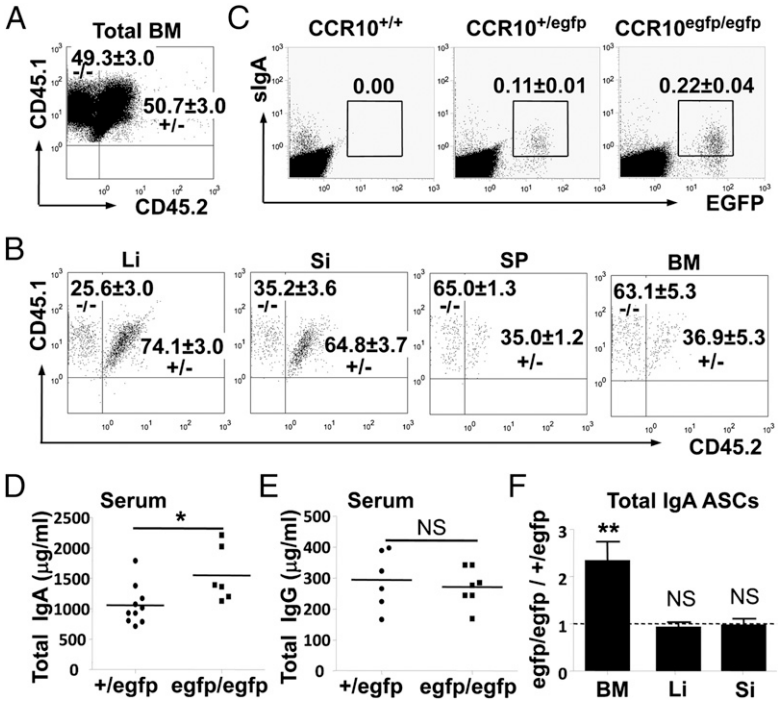

Fig. 4. Expression of CCR10 on IgA$^+$ cells is important for their proper maintenance in intestines. (*A* and *B*) Dysregulated distribution of CCR10-deficient IgA$^+$ cells in intestines and internal tissues. WT recipients (CD45.2$^+$CD45.1$^-$) reconstituted with the CCR10$^{EGFP/EGFP}$ (CD45.1$^+$CD45.2$^-$) and CCR10$^{+/EGFP}$ (CD45.1$^+$CD45.2$^+$) BM cells were analyzed for reconstitution of total BM cells (*A*) and EGFP$^+$IgA$^+$ cells (*B*) of CCR10$^{EGFP/EGFP}$ vs. CCR10$^{+/EGFP}$ origin in indicated tissues. Histographs gated on all donor-derived CD45.1$^+$ cells with percentages of CCR10$^{EGFP/EGFP}$ and CCR10$^{+/EGFP}$ origins indicated. SP, spleen; $n \geq 4$ mice per group. (*C*) Increased percentages of EGFP$^+$sIgA$^+$ cells in BM of CCR10$^{EGFP/EGFP}$ mice compared with CCR10$^{+/EGFP}$ controls, as assessed by flow cytometry; $n = 5$ mice per group. (*D* and *E*) Amounts of total IgA (*D*) and IgG (*E*) antibodies in sera of CCR10$^{EGFP/EGFP}$ and CCR10$^{+/EGFP}$ mice. One dot represents one mouse. (*F*) Relative ratios of total IgA-ASCs in BM and large and small intestines of CCR10$^{EGFP/EGFP}$ vs. CCR10$^{+/EGFP}$ mice. Numbers of total IgA-ASCs were assessed by using an ELISPOT assay. Ratios of the total IgA-ASC numbers in the indicated tissues of CCR10 $^{EGFP/EGFP}$ vs. CCR10 $^{+/EGFP}$ mice were presented, with 1 indicating no difference between the CCR10$^{EGFP/EGFP}$ and CCR10$^{+/EGFP}$ mice ($n = 6$ mice in each group).

and *E*). Relative abundance of monomeric and dimeric forms of IgA in sera of CCR10$^{EGFP/EGFP}$ and CCR10$^{+/EGFP}$ mice was similar (Fig. S9), consistent with a previous finding that the monomer is a dominant form of serum IgA, as the dimers (or polymers) could be transported into intestines through bile ducts even if they were generated in internal tissues and secreted into blood (20). The increased internal tissue-derived IgA antibodies likely also contribute to the intestinal IgA antibody pool. Taken together, our data demonstrate that absent expression of CCR10 on IgA$^+$ cells results in their dysregulated distribution in intestines and internal tissues, but enhanced generation of IgA$^+$ cells in ILFs and possibly other lymphoid tissues helps to offset the defect in intestines. Consistent with this, there were similar frequencies of total IgA antibody-secreting cells (IgA-ASCs) in intestines of CCR10$^{EGFP/EGFP}$ and CCR10$^{+/EGFP}$ mice based on an enzyme-linked immunosorbent spot (ELISPOT) assay, whereas higher frequencies of IgA-ASCs were found in BM cells of CCR10$^{EGFP/EGFP}$ than of CCR10$^{+/EGFP}$ mice (Fig. 4*F*).

Differential Effects of CCR10 KO on Intestinal IgA Responses to T-Dependent Antigen Stimulation and Bacterial Infection. To further understand the compensation process involved in intestinal IgA responses in CCR10$^{EGFP/EGFP}$ mice, we then tested how CCR10 KO affects intestinal IgA responses to T-dependent and -independent antigen stimulations, which should be different if the compensatory generation of IgA$^+$ cells is mainly through a T-independent process. Indeed, in a T-dependent model in which mice were orally challenged with the protein antigen CT, titers of CT-specific IgA antibodies were up to fivefold lower in CCR10$^{EGFP/EGFP}$ mice than in CCR10$^{+/EGFP}$ mice (Fig. 5*A*). In contrast, there was no difference in titers of fecal IgA specific to the infection in Rag1$^{-/-}$ mice that were reconstituted with naive CCR10$^{EGFP/EGFP}$ vs. CCR10$^{+/EGFP}$ B cells and orally infected with bacteria *Citrobacter rodentium*, a model of T-independent IgA responses (Fig. 5*B*) (21). These results support the notion that the compensatory generation of IgA in CCR10$^{EGFP/EGFP}$ mice is mainly through a T-independent process and is therefore not effective in response to T-dependent antigen stimulation. In a more natural but complicated setting, *Citrobacter*-infected CCR10$^{EGFP/EGFP}$ mice, which are capable of T-dependent and -independent IgA responses, had a slight delay in the *Citrobacter*-specific IgA production at an early time point after infection compared with the CCR10$^{+/EGFP}$ controls but caught up at all later time points (Fig. 5*C*) and did not have defects in clearance of the infection (Fig. S10), suggesting that any impaired intestinal migration of *Citrobacter*-specific IgA$^+$ cells was largely compensated. Consistent with the notion that CCR10-deficient IgA$^+$ plasma cells abnormally accumulate in internal tissues, serum levels of the *Citrobacter*-specific IgA, but not IgG, were higher in the infected CCR10$^{EGFP/EGFP}$ mice than in the CCR10$^{+/EGFP}$ mice at most time points, even long (i.e., > 3 mo) after the infection and clearance of the bacteria during the memory phase (Fig. 5*D* and Fig. S11).

Impaired Maintenance of Long-Lived *Citrobacter*-Specific IgA-Producing Plasma Cells in CCR10$^{EGFP/EGFP}$ Mice. Confirming the long-term accumulation of *Citrobacter*-specific IgA$^+$ plasma cells in internal tissues, higher frequencies of *Citrobacter*-specific IgA$^+$ spots were detected in BM cells of CCR10$^{EGFP/EGFP}$ mice than in those of CCR10$^{+/EGFP}$ mice 3 mo after their infection, as determined by an ELISPOT assay (Fig. 5*E*), Interestingly, frequencies of the *Citrobacter*-specific IgA$^+$ spots in LPL of the infected

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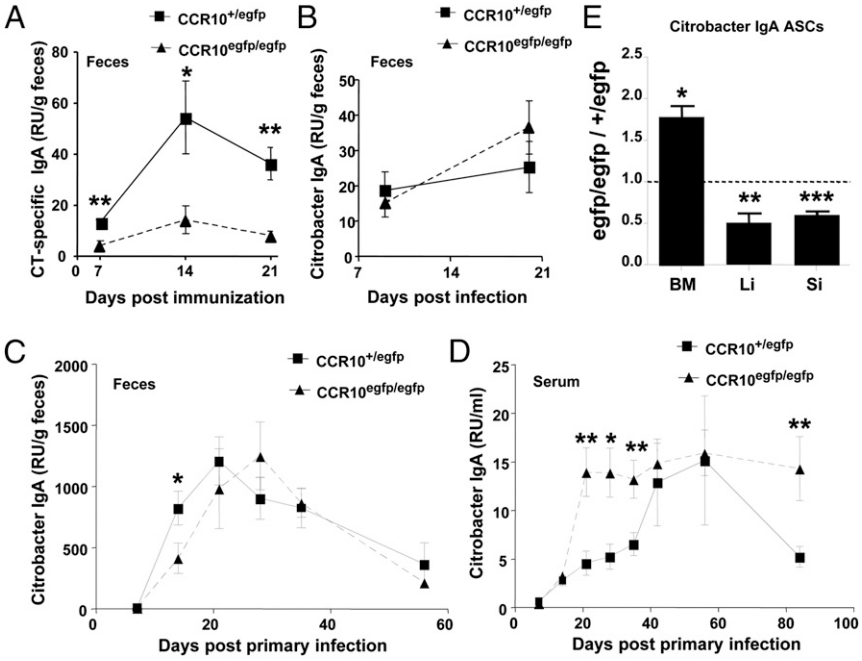

Fig. 5. Differential effects of CCR10 KO on the intestinal IgA response to T-dependent antigen stimulation and bacterial infection. (*A*) Titers of CT-specific IgA in feces of $CCR10^{EGFP/EGFP}$ and $CCR10^{+/EGFP}$ mice at different time points after oral challenge; n = 6–9 mice per group; RU, relative units (defined in *Methods*). (*B*) Titers of *Citrobacter*-specific IgA in feces of $Rag1^{-/-}$ mice transferred with naive $CCR10^{+/EGFP}$ or $CCR10^{EGFP/EGFP}$ B cells at days 7 and 14 after infection; n = 7 mice per group. (*C* and *D*) Titers of *Citrobacter*-specific IgA in feces and sera of $CCR10^{EGFP/EGFP}$ and $CCR10^{+/EGFP}$ mice at different time points after infection; n = 7–10 mice per group. (*E*) Relative ratios of *Citrobacter*-specific IgA-ASCs in BM cells and LPLs isolated from $CCR10^{EGFP/EGFP}$ vs. $CCR10^{+/EGFP}$ mice 3 mo after infection. The *Citrobacter*-specific IgA-ASCs were assessed by using an ELISPOT assay. The results were presented in the same way as for the total IgA-ASCs in Fig. 4*F*; n = 6 in each group.

$CCR10^{EGFP/EGFP}$ mice were significantly lower than in the $CCR10^{+/EGFP}$ controls 3 mo after infection (Fig. 5*E*), revealing an important role of CCR10 for the long-term maintenance of IgA-producing plasma cells in intestines. Therefore, although the defective intestinal migration of CCR10-deficient IgA^+ cells could be compensated for under homeostatic conditions or in the early phase of infection by the enhanced generation of IgA^+ cells as a result of stimulation from presence of the commensal or pathogenic bacteria, such a mechanism did not exist to offset the impaired maintenance of the long-lived *Citrobacter*-specific IgA-producing plasma cells after the bacteria are cleared.

Severely Impaired Memory IgA Response to *Citrobacter* Infection in $CCR10^{EGFP/EGFP}$ Mice. The impaired maintenance of long-lived *Citrobacter*-specific IgA^+ plasma cells in $CCR10^{EGFP/EGFP}$ mice might affect memory responses to reinfection of the bacteria. To test this, we reinfected the *Citrobacter*-infected $CCR10^{EGFP/EGFP}$ and $CCR10^{+/EGFP}$ mice 5 mo after their infection and clearance. In striking contrast to the generally normal primary IgA response, the memory IgA response to the *Citrobacter* infection in $CCR10^{EGFP/EGFP}$ mice was severely impaired (Fig. 6*A*). Two weeks after the reinfection, the $CCR10^{+/EGFP}$ mice reached the peak level of memory response, with the titer of *Citrobacter*-specific IgA in feces four times higher than that of the $CCR10^{EGFP/EGFP}$ mice (Fig. 6*A*). Four weeks after the reinfection, the fecal IgA level went down from its peak in the $CCR10^{+/EGFP}$ mice but kept increasing in the $CCR10^{EGFP/EGFP}$ mice, resulting in their comparable IgA levels at this time point (Fig. 6*A*). However, the IgA production in $CCR10^{EGFP/EGFP}$ mice never reached the peak level of $CCR10^{+/EGFP}$ mice. In addition, the significant difference in the fecal IgA level between the $CCR10^{EGFP/EGFP}$ and $CCR10^{+/EGFP}$ mice reappeared by end of the week 5 and thereafter because the $CCR10^{+/EGFP}$ mice maintained a high level of the *Citrobacter*-specific IgA for a long time but the $CCR10^{EGFP/EGFP}$ mice showed a rapid decrease (Fig. 6*A*). On the whole, the fecal IgA production pattern of the memory response in $CCR10^{EGFP/EGFP}$ mice was not significantly different from that of the primary response, whereas the IgA memory response was much faster and lasted much longer than the primary response in the $CCR10^{+/EGFP}$ mice (compare Fig. 5*C* vs. Fig. 6*A*). This demonstrates that $CCR10^{EGFP/EGFP}$ mice are almost devoid of the IgA memory response.

By using an ELISPOT assay, we confirmed that the severely impaired production of fecal *Citrobacter*-specific IgA antibodies in the reinfected $CCR10^{EGFP/EGFP}$ mice was a result of reduced numbers of *Citrobacter*-specific IgA-secreting plasma cells in the intestines. At day 14 after the reinfection, numbers of *Citrobacter*-specific IgA-secreting plasma cells in large intestines of $CCR10^{EGFP/EGFP}$ mice was drastically (15 fold) lower than in $CCR10^{+/EGFP}$ controls (Fig. 6*B*, *Top*). The numbers were also significantly (threefold) reduced in small intestines of the $CCR10^{EGFP/EGFP}$ mice (Fig. 6*B*, *Bottom*). The different extents of effect of CCR10 deficiency in the small and large intestines are likely caused by differential expression of other chemokine molecules involved in the migration of IgA^+ cells in these tissues, such as the CCR9 ligand CCL25 that is expressed only in the small intestine (9).

As seen in the primary response, there were much higher levels of *Citrobacter*-specific IgA in blood of the reinfected $CCR10^{EGFP/EGFP}$ mice than in $CCR10^{+/EGFP}$ mice most of the time (Fig. 6*C*). During a period between 1 and 2 mo after reinfection, serum levels of the *Citrobacter*-specific IgA were similar in the $CCR10^{+/EGFP}$ and $CCR10^{EGFP/EGFP}$ mice (Fig. 6*C*), likely because of the higher production of IgA in intestines of the $CCR10^{+/EGFP}$ mice (Fig. 6*A*). However, the difference in titers of serum IgA between $CCR10^{EGFP/EGFP}$ and $CCR10^{+/EGFP}$ mice reappeared after the intestinal IgA production went down (>3 mo past reinfection; Fig. 6*C*).

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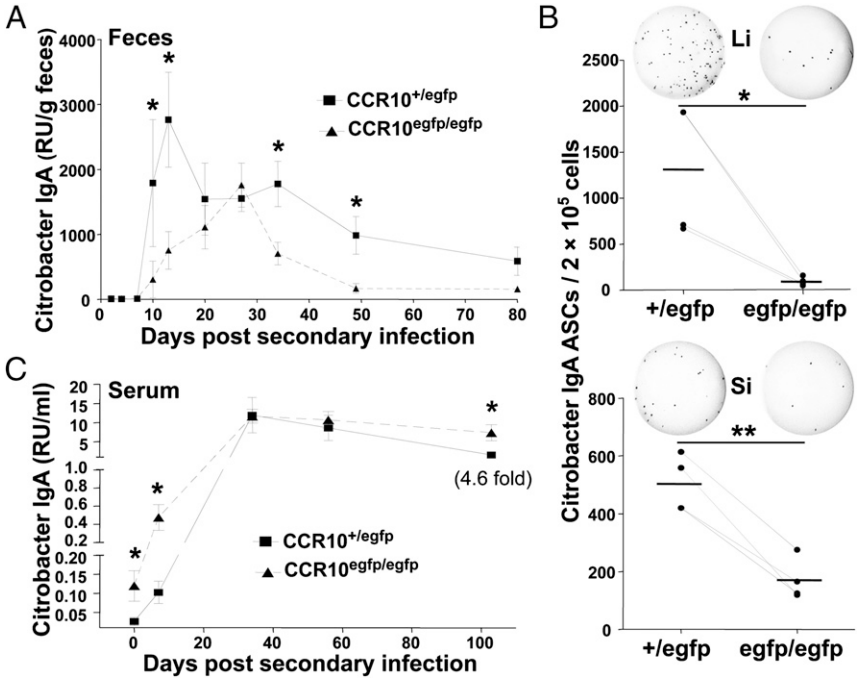

Fig. 6. Severely impaired intestinal IgA memory responses to the *Citrobacter* reinfection in CCR10-KO mice. (*A*) Titers of *Citrobacter*-specific IgA in feces of $CCR10^{EGFP/EGFP}$ and $CCR10^{+/EGFP}$ mice at different time points after reinfection; n = 7–10 for each group. (*B*) Frequencies of *Citrobacter*-specific IgA ASC in large and small intestinal LPLs of $CCR10^{EGFP/EGFP}$ and $CCR10^{+/EGFP}$ mice 14 d after the reinfection. The *Citrobacter*-specific IgA ASC was assessed using the ELISPOT assay. Microscopic pictures of representative wells of the ELISPOT are shown for each test group. The short flat lines indicate mean values of the groups. (*C*) Titers of *Citrobacter*-specific IgA in sera of $CCR10^{EGFP/EGFP}$ and $CCR10^{+/EGFP}$ mice at different time points after *Citrobacter* reinfection; n = 7–10 in each group.

Impaired Maintenance of *Citrobacter*-Specific IgA^+ Memory B Cells in Intestines of $CCR10^{EGFP/EGFP}$ Mice. The severely impaired IgA memory response in $CCR10^{EGFP/EGFP}$ mice suggests that any possible compensational generation of IgA^+ cells could not offset the impaired migration and maintenance of CCR10-deficient IgA-producing plasma cells during the memory response, likely because generation of IgA^+ cells from naive B cells in ILFs would require a much longer time than from the IgA^+ memory B cells. In addition, it is possible that maintenance of the IgA^+ memory B cells in infected $CCR10^{EGFP/EGFP}$ mice is also defective even before the reinfection. Supporting this possibility, CCR10 (EGFP) was expressed on a fraction of IgA^+ memory-like B cells ($CD19^+CD38^{hi}CD138^-$) of intestines, but not of other internal lymphoid tissues such as BM and MLN and blood (Fig. 7*A*) (22–24). Although frequencies of total $CCR10^+$ and $CCR10^-$ IgA^+ memory-like B cells in intestines were not significantly different in $CCR10^{EGFP/EGFP}$ vs. $CCR10^{+/EGFP}$ mice (Fig. S12), this could be because the effect of CCR10 KO on the memory cells is masked under homeostatic conditions, as with the IgA^+ plasma cells. Indeed, when *Citrobacter*-specific IgA^+ memory B cells were assessed by using an ELISPOT assay in $CCR10^{EGFP/EGFP}$ and $CCR10^{+/EGFP}$ mice 3 mo after their infection and clearance of *Citrobacter*, the *Citrobacter*-specific IgA^+ memory B-cell numbers were significantly lower in intestines of the $CCR10^{EGFP/EGFP}$ mice (Fig. 7*B*), whereas the ELISPOT assay still detected the same frequencies of total IgA^+ memory B cells in intestines of $CCR10^{EGFP/EGFP}$ and $CCR10^{+/EGFP}$ mice (Fig. 7*C*). $CCR10^{EGFP/EGFP}$ and $CCR10^{+/EGFP}$ mice also had similar frequencies of total and *Citrobacter*-specific IgA^+ memory B cells in their BM (Fig. 7 *B* and *C*), indicating that CCR10 KO specifically impaired intestinal maintenance of the *Citrobacter*-specific IgA^+ memory B cells.

Discussion

Proper long-term maintenance and memory responses of IgA production are important for protecting against pathogens that infect the mucosa. Understanding the molecular mechanisms that regulate these processes is critical for the design of better vaccines against many important intestinal pathogens, which has not been very successful. We report herein that CCR10 plays a critical role in these processes by regulating migration and maintenance of IgA^+ plasma cells as well as memory B cells. Our studies also reveal a compensational process that generates IgA^+ cells in ILFs of $CCR10^{EGFP/EGFP}$ mice to offset the impaired intestinal migration of IgA^+ cells. Although the compensatory process is sufficient in maintaining a normal level of the IgA production in $CCR10^{EGFP/EGFP}$ mice, quality/repertoire of the IgA antibodies are different from those of WT mice, and the compensation process could not rescue the impaired IgA memory response resulting from the CCR10 deficiency.

CCR10 is involved in the regulation of IgA memory responses in at least three aspects. First, it is important in maintenance of the long-lived IgA^+ plasma cells in the intestines. Second, it is required for migration of newly generated pathogen-specific IgA^+ plasma cells into the intestinal LP during the reinfection if the IgA^+ cells are not generated in situ. Third, CCR10 is important in the maintenance of IgA^+ memory B cells in the intestines. In $CCR10^{EGFP/EGFP}$ mice, all these three aspects of IgA memory responses were affected, resulting in the absence of memory IgA response to pathogen infection. How the respective effects of CCR10 deficiency on these three aspects affect the memory IgA response remains to be determined.

It is interesting that only a fraction of intestinal IgA^+ memory-like B cells express CCR10. This suggests that this intestine-specific population is different from the other IgA^+ memory B-cell populations of intestines or other lymphoid tissues. One possibility is that this population might represent the intestine-specific ef-

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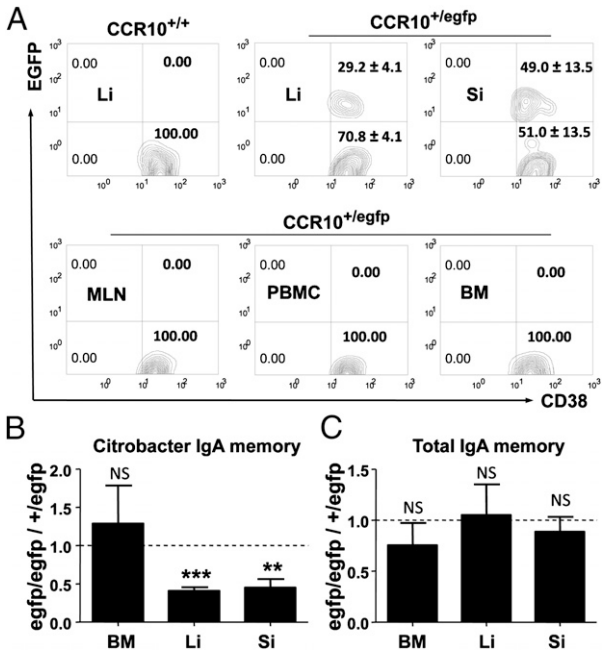

Fig. 7. Impaired maintenance of *Citrobacter*-specific IgA$^+$ memory B cells in intestines of CCR10-KO mice. (*A*) Unique expression of CCR10 on intestinal IgA$^+$ memory-like B cells. Lymphocytes of large and small intestines, MLN, peripheral blood [i.e., peripheral blood mononuclear cell (PBMC)], and BM of CCR10$^{+/EGFP}$ mice were analyzed by FACS for CD19, CD38, CD138, and sIgA expression. Graphs are of gated sIgA$^+$CD19^{hi}CD38^{hi}CD138$^-$ cells. The intestinal cells of WT mice (*Top Left*) were included as negative controls for EGFP expression; n = 3–5 in each group. (*B* and *C*) Relative ratios of total and *Citrobacter*-specific IgA$^+$ memory B cells in intestines and BM of CCR10$^{EGFP/EGFP}$ vs. CCR10$^{+/EGFP}$ mice. For detection of *Citrobacter*-specific IgA$^+$ memory B cells (*B*), mice infected with *Citrobacter* 3 mo earlier were used, whereas naive mice were used to determine total IgA$^+$ memory B cells (*C*) by a method in which total B cells were cultured in vitro for 6 d in presence of LPS, pokeweed mitogen, CpG, and IL-2 before being used on the total or *Citrobacter*-specific IgA ELISPOT assay. Ratios of the IgA$^+$ memory B cells in intestines and BM of CCR10 $^{EGFP/EGFP}$ vs. CCR10 $^{+/EGFP}$ mice were presented, with 1 indicating no difference between them (n = 5 mice in each group).

fector memory B cells, whereas the others are more like central memory B cells, analogues to the division of effector and central memory T cells (25). Although further studies are required to determine whether this is the case, it is clear that CCR10 is required in the maintenance of IgA$^+$ memory B cells specifically in the intestines, which correlates well with the specific CCR10 expression by the intestinal IgA$^+$ memory-like B cells.

The relationship between the CCR10$^-$ and CCR10$^+$ memory B-cell populations will be an important question to understand how the pathogen-specific IgA memory is maintained and activated to generate the IgA-producing plasma cells rapidly. One possibility is that the CCR10$^+$ B-cell population is derived from the CCR10$^-$ memory B-cell population in the unique environment of intestines. Considering that the CCR10$^+$ memory-like B cells are more similar to the CCR10$^+$ IgA-producing plasma cells, they might represent a population of a transition stage during the differentiation of the CCR10$^-$ IgA$^+$ memory B cells into plasma cells. However, it is possible that the two populations are generated independently during the memory B-cell formation.

The CCR10-KO mice are not only devoid of the ability of rapid generation of IgA antibodies characterizing the memory response, they are also defective in the long-term maintenance of IgA production in the intestine, particularly during the IgA memory response to the bacteria reinfection. This suggests that CCR10 might play an important role in survival/proliferation of the long-lived IgA-producing plasma cells, likely by transducing the CCL28-initiated signals. It was recently found that IgA- producing plasma cells could be maintained in the intestinal environment for a long time without stimulation from antigens (8). Our findings suggest that the CCR10-transduced signals might contribute to the long-term maintenance of IgA-producing plasma cells in the intestines. Although our antibiotic treatment experiment here did not reduce the total fecal IgA level in CCR10-KO mice, this could be a result of the short duration of the treatment and/or continuing presence of commensal bacteria. Testing how CCR10-deficient IgA$^+$ cells are maintained in absence of antigen stimulation, such as in germ-free mice, might provide a "cleaner" model to determine involvement of CCR10 in IgA$^+$ cell maintenance. Probably relevant to this, it was noted that CCR10-transduced signals promote survival/proliferation of CCR10-expressing melanoma cells in the skin (26), an epithelial site that highly expresses CCL27, the other ligand of CCR10 (27).

Although the compensational generation of IgA$^+$ cells in ILFs of CCR10$^{EGFP/EGFP}$ mice is able to offset the impaired migration and maintenance of IgA$^+$ cells to provide a normal level of IgA production under homeostatic conditions and in a primary response to the infection, the quality of IgA antibodies generated from the compensational process might be different from those of WT mice, as it is known that the generation of IgA$^+$ cells in ILFs is mainly through a T-cell–independent process (3), which generally gives rise to IgA antibodies of less diversity and lower affinity than those generated through a T-cell–dependent process, as in germinal centers of Peyer patches. Indeed, intestinal IgA$^+$ cells of CCR10$^{EGFP/EGFP}$ mice had few mutations in their IgH alleles, and IgA responses to T-dependent antigens could not be efficiently compensated for in CCR10$^{EGFP/EGFP}$ mice. In addition, the compensational process is dynamic, depending on conditions of intestinal environments. Likely, the compensational process would be most active in young mice during the initial stage of bacterial colonization of intestines. When the IgA levels in the intestine are sufficient to maintain homeostasis of commensal microbes, the enhanced generation of IgA$^+$ cells might not be necessary and could be reduced. However, when the balance has been disrupted, as in the case of infection, the compensational generation of IgA$^+$ cells might be reactivated. Although this is efficient in providing the primary IgA response to the bacteria infection, it is clear that any compensatory generation of IgA$^+$ cells from naive B cells could not rescue the multiple defects in IgA memory responses in CCR10$^{EGFP/EGFP}$ mice.

Our findings could aid in the designing of vaccines against many important pathogens that infect the intestines and other mucosal sites, such as lungs and reproductive tracts, where IgA antibodies play an important role. Recent studies found that systemic immunization of animals with the CCR10 ligand CCL27 or CCL28 as an adjuvant increased titers of antigen-specific IgA antibodies in mucosal tissues such as intestines, suggesting that manipulating the CCR10/ligand axis could be useful in enhancing the vaccination efficacy against mucosal pathogens (28, 29). It will be interesting to study how CCR10/ligands increase the mucosal IgA memory in this setting.

Methods

Mice. CCR10-KO/EGFP-knock-in mice on C57BL/6 (B6) background (CD45.2$^+$) were described previously (17), and crossed to B6 mice bearing CD45.1 alleles (Jackson Laboratory) to obtain CCR10$^{EGFP/EGFP}$ and CCR10$^{+/EGFP}$ mice bearing CD45.1 and/or CD45.2 alleles. Rag1$^{-/-}$ mice were from Jackson Laboratory. Mice were used at 8 to 12 wk of age unless indicated otherwise. All mouse experiments were performed in specific pathogen-free conditions in accordance with protocols approved by the Pennsylvania State University Institutional Animal Care and Use Committee.

Antibodies and Chemicals. PE-Cy7–conjugated anti-CD45.1 (A20), PE-Cy7–conjugated anti-CD3ε (145-2c11), biotin-conjugated anti-IgA (11-44-2), purified or Alexa Fluor 647-conjugated anti-B220 (RA3-6B2), PE- or PE-Cy7–

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conjugated anti-CD19 (1D3), and PE-Cy5–conjugated anti-CD38 (clone 90) antibodies were purchased from eBioscience. APC-conjugated anti-CD45.2 (clone 104), biotin-conjugated anti-CD138 (281-2), purified rat anti-mouse CD16/CD32 (2.4G2) antibodies, and PE-Texas Red–conjugated streptavidin were from BD Bioscience. PE-conjugated anti-IgA (11-44-2) was from Southern Biotech. Alexa Fluor 568-conjugated streptavidin was from Molecular Probes. Peroxidase-conjugated AffiniPure goat anti-rat IgG (H + L) was from Jackson ImmunoResearch. CT was from List Biological.

BM Transfer. Cell sorter-isolated EGFP$^-$ BM cells of CCR10$^{+/EGFP}$ (CD45.1$^+$CD45.2$^+$) and CCD10$^{EGFP/EGFP}$ (CD45.1$^+$CD45.2$^-$) mice were 1:1 mixed and injected i.v. into lethally irradiated (950 rad) WT B6 (CD45.1$^-$D45.2$^+$) mice (total 10^6 cells per mouse). The recipients were analyzed 7 to 8 wk after the transfer.

Oral Immunization with CT. On days 1 and 3, mice were deprived from food for 8 h and gavaged with two doses of CT (10 μg in 0.5 mL PBS solution containing 3% sodium bicarbonate), and analyzed on 7, 14, and 21 d after the first immunization.

***Citrobacter* Infection.** Mice were fasted for 8 h and then orally infected by gavage with 2×10^9 cfu of *C. rodentium* strain DBS100 (ATCC 51459; American Type Culture Collection) in a total volume of 200 μL per mouse in both the primary and secondary infection. The secondary infection was performed on mice that were infected with *Citrobacter* 5 mo earlier and confirmed cleared of the infection. To assess T-independent IgA responses, Rag1$^{-/-}$ mice were transferred with 10^6 EGFP$^-$B220$^+$IgM$^+$ B cells of naive CCR10$^{+/EGFP}$ or CCR10$^{EGFP/EGFP}$ mice 1 d before they were infected with *Citrobacter*. The cfu count of the bacteria used in the infection was estimated by absorbance at the wavelength of 600 nm and confirmed by serially diluting and plating the inoculums on MacConkey agar plates (Becton Dickinson). For quantification of bacterial titers after the infection, feces were homogenized in PBS solution, serially diluted, and plated on MacConkey agar plates. Colonies formed were counted after an overnight culture.

Antibiotic Treatment. Mice were fed with an antibiotic mixture of 500 mg/L ampicillin (Sigma), 500 mg/L vancomycin (Sigma), 1 g/L neomycin (USB), and 1 g/L metronidazole (Sigma) in drinking water as reported previously (30).

Assessment of Somatic Hypermutation Rates in IgH Alleles. Genomic DNA were prepared from cell sorter-purified intestinal EGFP$^+$sIgA$^+$ cells or B220$^+$IgM$^+$ spleen cells and analyzed for the hypermutation frequencies in a 3′ region franking V_HJ558/D/J_H4 rearrangements as previously described (31).

Quantification of Commensal Bacteria. Quantification of commensal bacteria was performed as previously described (32, 33).

Western Blot Quantification of Monomeric and Dimeric IgA Antibodies. Serum samples were run on 6% nonreducing SDS/PAGE gel, transferred to nitrocellucose, blotted with alkaline phosphatase-conjugated goat anti-mouse IgA (Santa Cruz Biotechnology), and developed with the enhanced chemiluminescence substrate (GE Healthcare). Fluorescent signals from products of the enhanced chemiluminescence substrate were detected by using the FLA-7000 system (Fujifilm), and intensities of monomeric and dimeric IgA bands were quantified by using Multi gauge software.

Whole-Mount Staining of Intestines for Detection of B220$^+$ Clusters of ILFs. Whole-mount staining of intestines for detection of B220$^+$ clusters of ILFs was performed as previously reported (18, 19).

Immunofluorescent Microscopy. Intestines were flushed with PBS solution, opened longitudinally along the mesenteric border, and fixed for 2 h at 4 °C in a fresh 4% paraformaldehyde solution. After three washes in PBS solution, the tissues were incubated in a 30% sucrose solution overnight at 4 °C, and then frozen in OCT compound (Sakura Finetek). The frozen tissues were cut into 6-μm sections and stained with fluorescently labeled antibodies in an antibody amplifier (ProHisto) according to the manufacturer's instructions. The stained sections were covered with mounting medium containing DAPI (Vector) and examined under a FluoView FV1000 confocal microscope (Olympus). Images were processed with FV10-ASW software (Olympus).

In Vivo Migration Assay. Intestinal LP lymphocytes were isolated from CCR10$^{+/EGFP}$ (CD45.1$^+$CD45.2$^+$) and CCD10$^{EGFP/EGFP}$ (CD45.1$^-$CD45.2$^+$) mice. Similar numbers of EGFP$^+$sIgA$^+$ cells of the two kinds of mice were mixed and injected i.v. into C57BL/6 WT recipient mice (CD45.2$^-$CD45.2$^+$). Sixty hours after the injection, lymphocytes were isolated from large and small intestines of the recipients separately and FACS analyzed for EGFP, sIgA, CD45.1, and CD45.2 expression. Percentages of the CCR10$^{+/EGFP}$ vs. CCR10$^{EGFP/EGFP}$ donor cells in gated EGFP$^+$sIgA$^+$ populations were determined based on the CD45.1 and CD45.2 expression.

Cell Isolation. LP lymphocytes were isolated from large and small intestines as described (34), with some modifications. In brief, intestines were flushed with cold HBSS containing 15 mM Hepes (CMF/Hepes). After removal of fat, connective tissues, and Peyer patches, the intestines were opened longitudinally, cut into 5-mm segments, and washed four or five times with cold CMF/Hepes. The intestinal segments were then incubated for 15 min at 37 °C with shaking on a shaker (Model: GyromaxTM 737, Amerex Instrument Inc., Lafayette, CA) at 200 rpm in HBSS containing 15 mM Hepes, 5 mM EDTA, and 10% FBS (Equitech-Bio), followed by intense vortexing to remove the epithelium and intraepithelial lymphocytes. This step was repeated three or four times until no more epithelium shedding occurred. The remaining pieces were washed with RPMI medium (Mediatech) containing 10% FBS, minced, and then digested for 60 min at 37 °C with shaking at 200 rpm in RPMI medium containing 5% FBS, 0.6 mg/mL collagenase (Worthington), and penicillin and streptomycin (Gibco). Dissociated cells from the digestion were washed once with PBS solution, resuspended in 5 mL of 100% Percoll (GE Healthcare), underlaid with 4 mL of 40% Percoll, and centrifuged for 20 min at $850 \times g$ at room temperature. LP lymphocytes were recovered from the interphase of the Percoll gradient, washed twice, and resuspended in FACS buffer (PBS solution containing 3% FBS and 0.05% sodium azide) or RPMI medium. To isolate lymphocytes from Peyer patches, Peyer patches were gently homogenized with a tissue homogenizer, passed through a 70-μm cell strainer (BD Biosciences), and enriched by Percoll gradient centrifugation. BM cells were isolated by flushing tibias and femurs. Spleen and MLN cells were prepared by pressing the tissues through cell strainers using the end of a sterile plunger of a 5-mL syringe. Peripheral blood lymphocytes were isolated by gradient centrifugation by using Lympholyte-Mammal (Cedarlane Laboratories).

Flow Cytometry. All FACS analyses were performed on an FC500 series system (Beckman Coulter), and data were analyzed with FlowJo software (TreeStar).

ELISA. Samples of feces and blood were analyzed for the total and *Citrobacter*-specific IgA or IgG antibodies. Feces were collected and homogenized in 1 mL of TBS–BSA solution (50 mM Tris, 140 mM NaCl, 1% BSA, pH 8.0) per 100 mg feces. After centrifugation at $16,000 \times g$ for 5 min, the supernatants were collected and kept at −80 °C until the ELISA. Blood samples were collected by cheek puncture, put at room temperature for 30 min and 4 °C for 2 h, and centrifuged at $2,500 \times g$ for 10 min. Resulting sera were collected after the centrifugation and kept at −80 °C until analysis.

Total IgA and IgG levels in the fecal and serum samples were determined by using a Mouse IgA ELISA Quantitation Set and Mouse IgG ELISA Quantitation Set (Bethyl Laboratories), respectively, following the manufacturer's instructions.

For quantification of bacteria-specific IgA or IgG antibodies, 96-well ELISA plates (Costar) were coated overnight at 4 °C with 100 μL of soluble proteins (10 μg/mL in PBS solution) prepared from centrifugation supernatants ($16,000 \times g$, 1 h at 4 °C) of sonicated *C. rodentium, Clostridium perfrigens* (ATCC 13124D-5), *Eubacterium rectale* (ATCC 33656), *Lactococcus lactis* (ATCC 11454), or *Escherichia coli* (ATCC 43890) (24 cycles of 5 s burst plus 10 s resting, on ice). The wells were then washed five times and blocked for 2 h at room temperature with 200 μL of TBS–BSA solution, followed with two washes again. Wells were then incubated for 1 h at room temperature with 100 μL of samples diluted in the TBST–BSA solution. Wells were again washed five times, and incubated for 1 h at room temperature with HRP-conjugated anti-mouse IgA or HRP-conjugated anti-mouse IgG (Bethyl Laboratories). After five washes, wells were developed with 100 μL of TMB substrate solution (BD Biosciences) for 5 to 15 min, and then stopped with 100 μL of 2 N H_2SO_4 solution. The optical density was read at 450 nm. A set of twofold serially diluted fecal extracts or serum samples was always included to produce a standard curve in every experiment. The sample served as the standard was kept at −80 °C as aliquots to avoid freezing and thawing, and the same standard was used in every experiment. Relative concentrations of bacteria-specific IgA or IgG antibodies in tested samples were calculated by comparing to the standard curve and expressed as relative units.

ELISPOT. The frequency of total or *Citrobacter*-specific IgA–secreting plasma cells in intestinal lymphocyte preparations or BM cells was determined by

ELISPOT, similar as previously described (35, 36) with a minor modification. In brief, MultiScreen IP filtration plates (Millipore) were prewetted for 1 min with 15 μL of 35% ethanol, rinsed three times with sterile PBS solution, and then coated with 100 μg/mL of 1:50-diluted anti-IgA coating antibody (Bethyl Laboratories) or *C. rodentium* soluble proteins in PBS solution overnight at 4 °C, followed by five washes with PBS solution. The plates were blocked for 2 h at 37 °C with RPMI medium supplemented with 10% FBS, 50 μM β-mercaptoethanol, glutamine, and penicillin and streptomycin (RPMI-10). Cells to be tested were then transferred to the plates in serial dilutions and cultured overnight at 37 °C in the RPMI-10 medium. The plates were washed five times with PBS solution containing 0.01% Tween-20, and incubated for 1 h at 37 °C with 1:5,000 diluted HRP-conjugated anti-IgA detection antibody (Bethyl Laboratories), followed by five washes. The reactions were developed by using TMB substrate for ELISPOT assay (Mabtech) and stopped by washing under running water. Spots corresponding to antibody-secreting cells were counted under a dissecting microscope. The results were normalized to total cell inputs.

Detection of total or *Citrobacter*-specific memory B cells was also performed as previously described (35, 36) with minor modifications. Briefly, BM cells or intestinal lymphocyte preparations were cultured at 5×10^5 cells/mL and stimulated with 5 μg/mL PWM (Sigma-Aldrich), 6 μg/mL type B CpG (ODN-1668; InvivoGen), 2 μg/mL LPS (Sigma-Aldrich), and 100 U/mL IL-2 in the RPMI-10 medium for 6 d in 24-well plates. After stimulation, cells were washed three times with RPMI medium and used in the ELISPOT assays. Frequencies of the *Citrobacter*-specific IgA memory B cells were calculated from the numbers of IgA^+ spots normalized to the initial cell inputs.

Statistical Analyses. All data are expressed as means ± SEM. Statistical significance was determined by two-tailed Student *t* tests, with $P < 0.05$ as the threshold of significance.

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